

Major Project in Electrical Engineering

The V-Luxe



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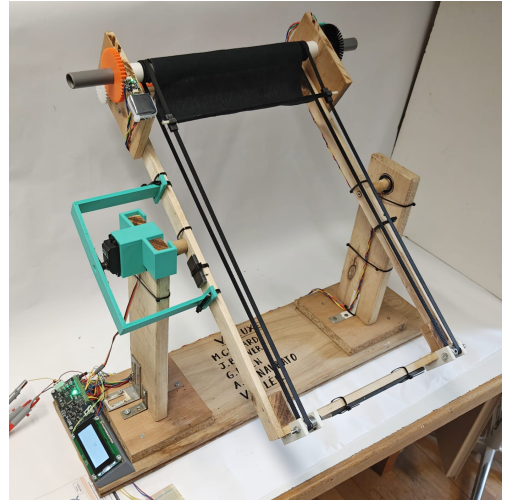
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EYEN

Matthieu
GILLARD

Arnaud
INNAURATO

Introduction

- Scale model of a tilting window
- Electronically controlled blind
- 2 modes of operation
- Sensor based system

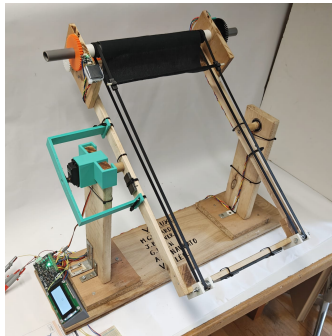


Overview

1. Mechanics
2. Electronics
3. Control System
4. Embedded Software
5. Management
6. Improvements

Mechanics

- Wooden structural frame, rotating on ball bearings
- PVC blind roller driven by gear system
- 3D-printed parts for:
 - Entire QEI system
 - Link between servo and window frame
 - PCB/Screen mount
 - Various gears and mounts
- Iron counterweight for balancing (reduces servo load)



Electronics: Microcontroller

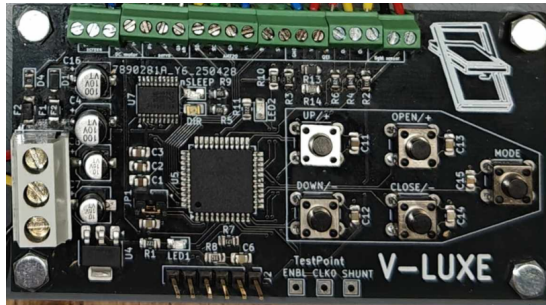
- **PIC18F4331** chosen for its native QEI module
- Capabilities used :
 - QEI
 - PWM generators
 - ADC channels
 - Digital I/O with pull-up resistance
 - Timers
- Lack of native I^2C master support $\implies$ application-specific design (time-consuming)

Electronics: Power delivery design

- Dual power sources
 - 9V for DC motor
 - 5.5V for servomotor (peak 1.4A $\Rightarrow$ separated to avoid regulator overload)
- 5V rail regulated from 9V using **LM1117** LDO :
 - Regulator-powered devices: PIC, LCD, sensors
- Protections at both power inputs:
 - Schottky diode
 - Fuse

Electronics: PCB

- 2-layer PCB with dedicated GND plane (KiCad 8.0)
- Terminal blocks to connect off-board components

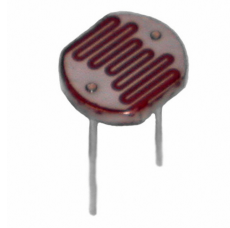


Electronics: Sensors overview

Temperature & Humidity AHT20



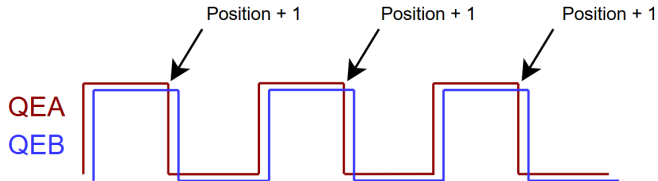
Light Sensor PDV-P5001



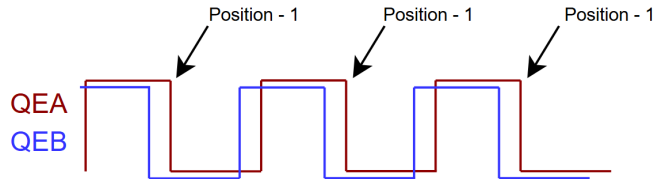
Optical Switches TCST2103



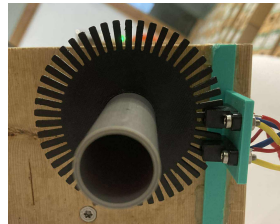
Quadrature Encoder Interface (QEI)



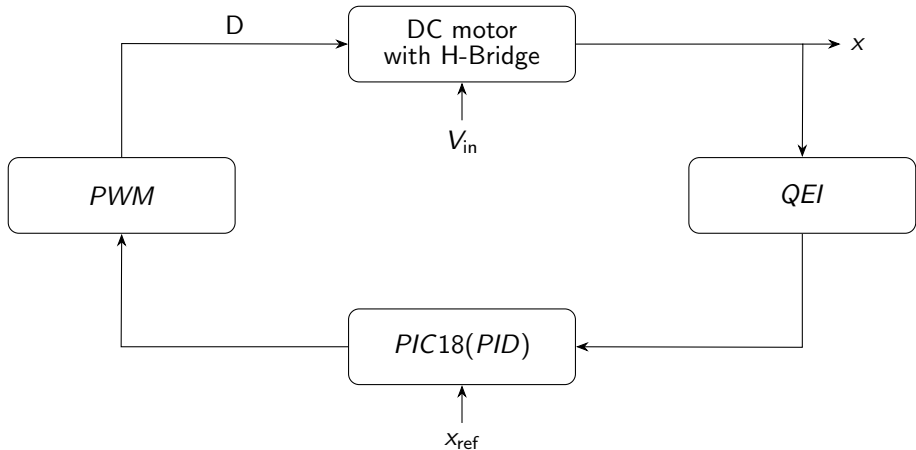
Clockwise



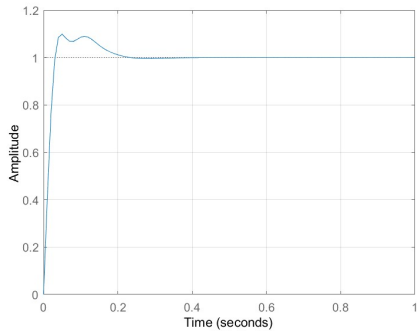
Counterclockwise



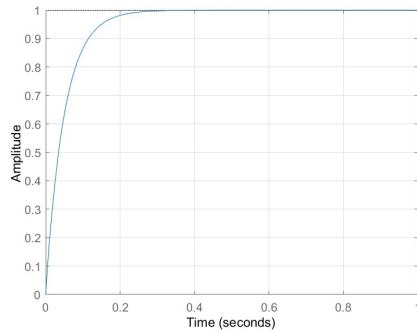
Control System: Loop



Control System: Design



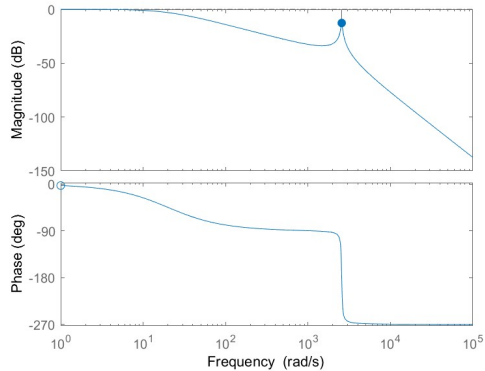
$$K_p = 90, K_i = 1500, K_d = 1.35$$



$$K_p = 25$$

Control System: Design

- Proportionnal controller
- $K_p = 25$
- Switching frequency = 50 *Hz*



Control System: Implementation

- Proportional control with $k_p = 25$
- Output of the controller encoded on 13-bits
- The duty cycle of the DC motor controlled with 7 bits
- Trade-off between range of control and position precision
- Overcome the static error with integrative action

Software: Tasks

Task	Description	Frequency
Temp/Hum	Command the I^2C communications	1 Hz
Luminosity	Get the exterior luminosity	1 Hz
Screen	Update the screen content	1 Hz
Servo	Update the servomotor PWM	1 Hz
Decision	Compute ideal state based on measurements	1 Hz
PID	Control the DC motor	50 Hz
Buttons	Check for the button inputs	50 Hz
I^2C	Perform the I^2C communications	3400 Hz

Software: General architecture

- Slow processes
- No need for preemption
- Interrupts
 - Round Robin with interrupts

Round-robin with interrupts

priority



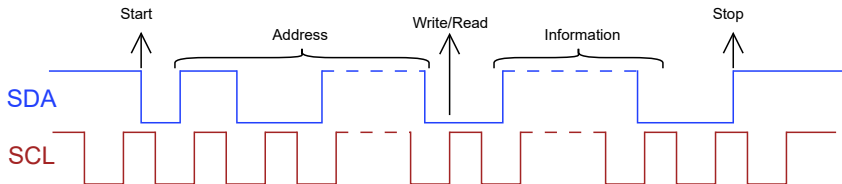
I2C display &
temperature/humidity
3400Hz

Buttons & PID
50Hz

Decison, Sensors &
Servo command
1Hz

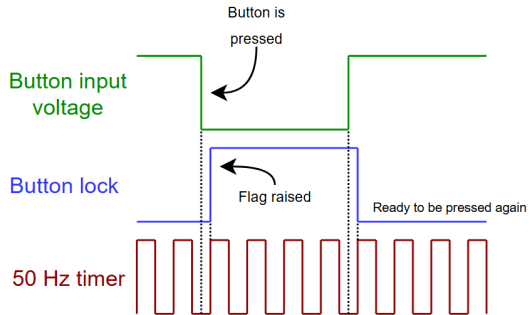
Software interrupt: I^2C

- One for the screen + one per TH sensor (grouped)
- Time sensitive
- Need to play with changing-direction pins
- Frequency of the interrupt = 3.4 Hz
- SCL frequency = 1.7 Hz



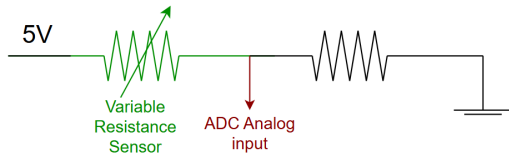
Software tasks: Buttons

- Digital I/O port
- Anti rebound capacitance
- Pull up resistances
- Single press detection
- 50 Hz



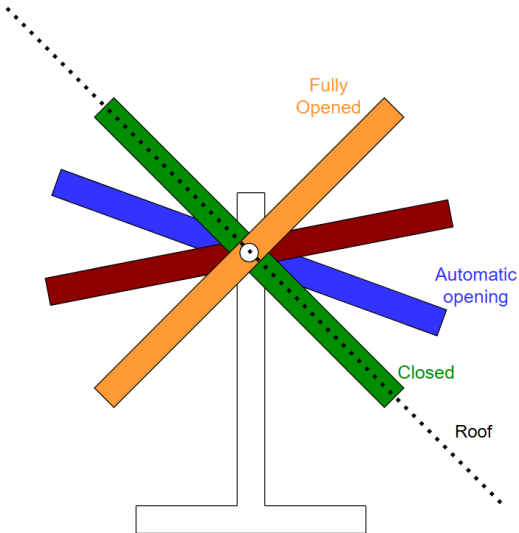
Software tasks: Luminosity

- ADC input
- “Sunny” or “Not sunny” threshold
- 1 Hz



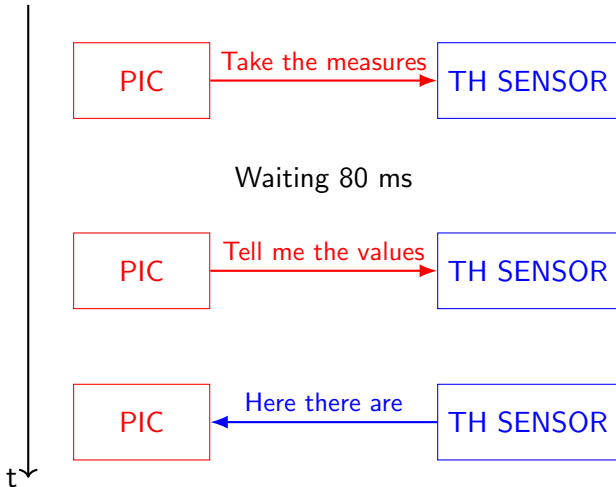
Software tasks: Servo

- PWM output
- Four possible states in manual
- Two in automatic
- 1 Hz



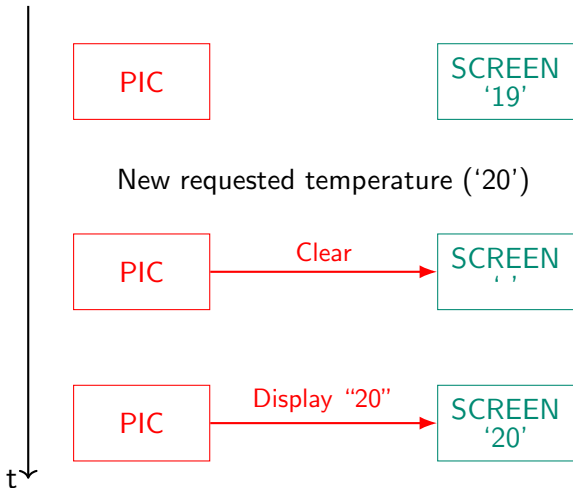
Software tasks: Temperature/Humidity

- Slow-changing variables
- Set I^2C mode: writing or reading
- Launch I^2C
- 1 Hz

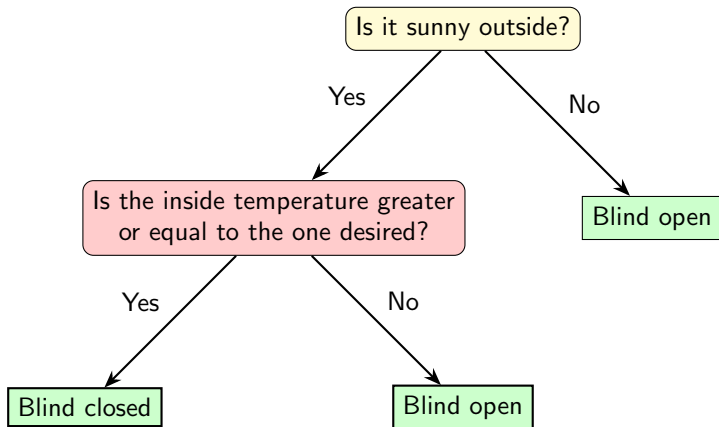


Software tasks: Screen

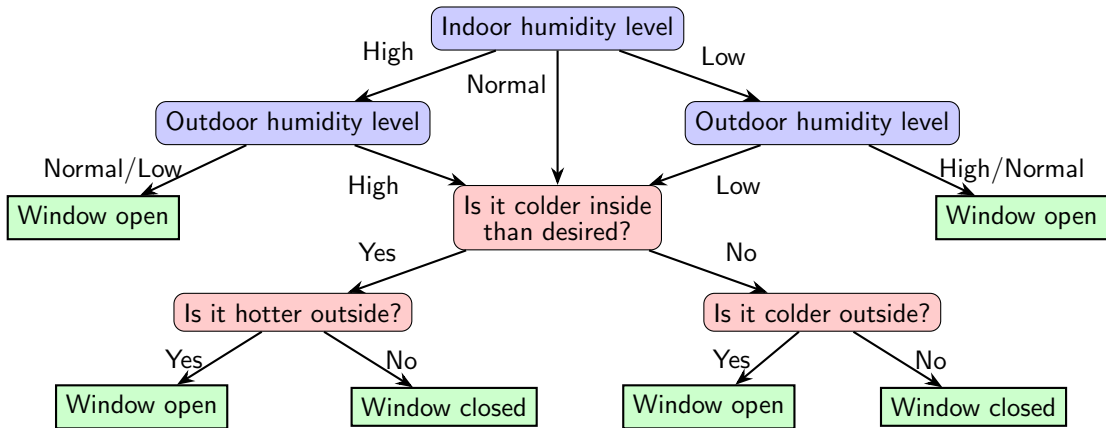
- If requested temperature is changed
- Load value + launch I^2C
- 1 Hz



Software tasks: Decision

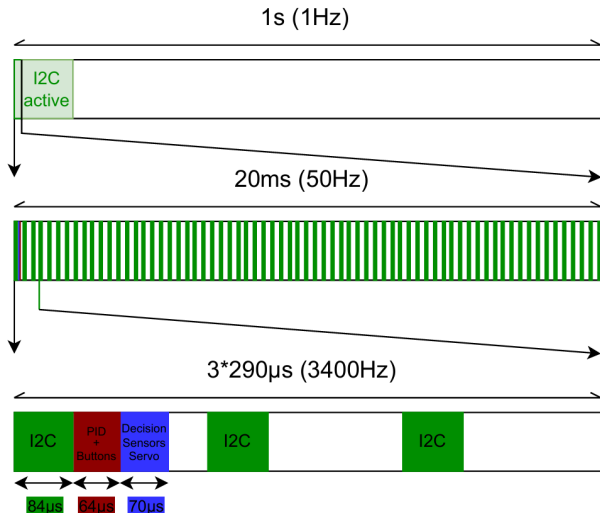


Software tasks: Decision



Software: Schedulability

- Slowest timer period $\approx 1\text{s}$
- I^2C active for $\approx 100\text{ms}$
- Processor load factor $\approx 3\%$



Management: Backward analysis

- Whole year project
- Importance of planning and team cohesion
- Strengths of all the members are different
- Take the time to debrief and reevaluate the tasks and planning

Management: Risk mitigation

Block	Problem	Mitigation
Power Supply	Bouncing signals → possible unstable voltage (RPN = 392)	Add fuse + voltage regulator
Microcontroller	Excess current → risk of damage (RPN = 192)	Add fuse
Security	Hand stuck → safety hazard (RPN = 126)	Servo not powerfull enough to harm
User interface	Button bounce → unwanted function (RPN = 105)	Add debounce

Focus: Risks with high severity

Goal: Reduce severity or probability through actions

Things that should have been done another way

- More complex electronics to acquire more skills in that topic
- Have carefully read specifications of components (I^2C master mode block in this PIC)
- Servomotor for window tilting induces oscillations that could have been avoided with DC motor

Any questions?

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We survived year one of the master's—cheers to that!